

Risk Parity ETF Design & Tracking Error Optimization

FIN 285A: Computer Simulations & Risk Assessment

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Project Overview

Part 1: Risk Parity ETF

Design a leveraged risk parity portfolio using four broad asset classes:

- AGG — Global Nominal Bonds
- ACWI — Global Equities
- GSG — Commodities
- TIP — Inflation-Linked Bonds

EWMA risk model ($\lambda=0.97$), 200% leverage,
18-month rolling window, monthly rebalancing

Part 2: ETF Optimization

Minimize tracking error with 20-asset expanded universe:

- Equity (7): VTI, HEDJ, VWO, QQQ, IWM, EWJ, IND
- Bonds (5): AGG, BNDX, TLT, LQD, HYG
- TIPS (2): TIP, STPZ
- Commodities (6): GLD, USO, CPER, SLV, DBA, UNG

Compare MA vs EWMA models,
train/test validation (2014–2018 / 2019–2025)

Part 1: Methodology & Benchmark

Risk Parity Framework

1. Equal risk contribution across all 4 asset classes
2. EWMA covariance ($\lambda = 0.97$, demeaned returns)
3. 18-month rolling window (tuned via grid search)
4. 200% leverage (weights sum to 2.0)
5. 10% minimum weight per asset
6. Monthly rebalancing after 5-year burn-in
7. Data: 2008–2024 (monthly from Yahoo Finance)

ALLW Benchmark (Rescaled to 200%)

Asset Class	Proxy	Weight
Global Nominal Bonds	AGG	76.12%
Global Equities	ACWI	48.49%
Commodities	GSG	40.66%
Inflation-Linked Bonds	TIP	34.74%
Total Notional		200.00%

Why These Benchmark Weights?

- Bonds get ~76% because they are ~3x less volatile than equities — equal risk contribution requires higher notional exposure
- Equities (~48%) need less capital to contribute the same risk
- Commodities & TIPS provide inflation protection at ~41% and ~35%
- Total = 200% → leverage amplifies returns while maintaining balanced risk

Part 1: Why These Specific ETF Proxies?

AGG — Global Nominal Bonds

Chosen over BND: AGG tracks the Bloomberg U.S. Aggregate Bond Index, identical coverage but higher AUM (\$110B+) and tighter spreads. Inception 2003 — provides 20+ years of data. Captures IG corporate, government, and MBS bonds.

ACWI — Global Equities

Chosen over VT: ACWI tracks the MSCI All Country World Index — covers 23 developed + 24 emerging markets in a single ETF. Inception 2008 — aligns with our data start. Better match to Bridgewater's global equity mandate than U.S.-only alternatives like SPY.

GSG — Commodities

Chosen over DBC: GSG tracks the S&P GSCI — broader coverage across energy, agriculture, industrial metals, precious metals, and livestock. DBC only tracks 14 commodities vs GSG's 24. Inception 2006 — longer history than most commodity ETFs. Acts as natural inflation hedge.

TIP — Inflation-Linked Bonds

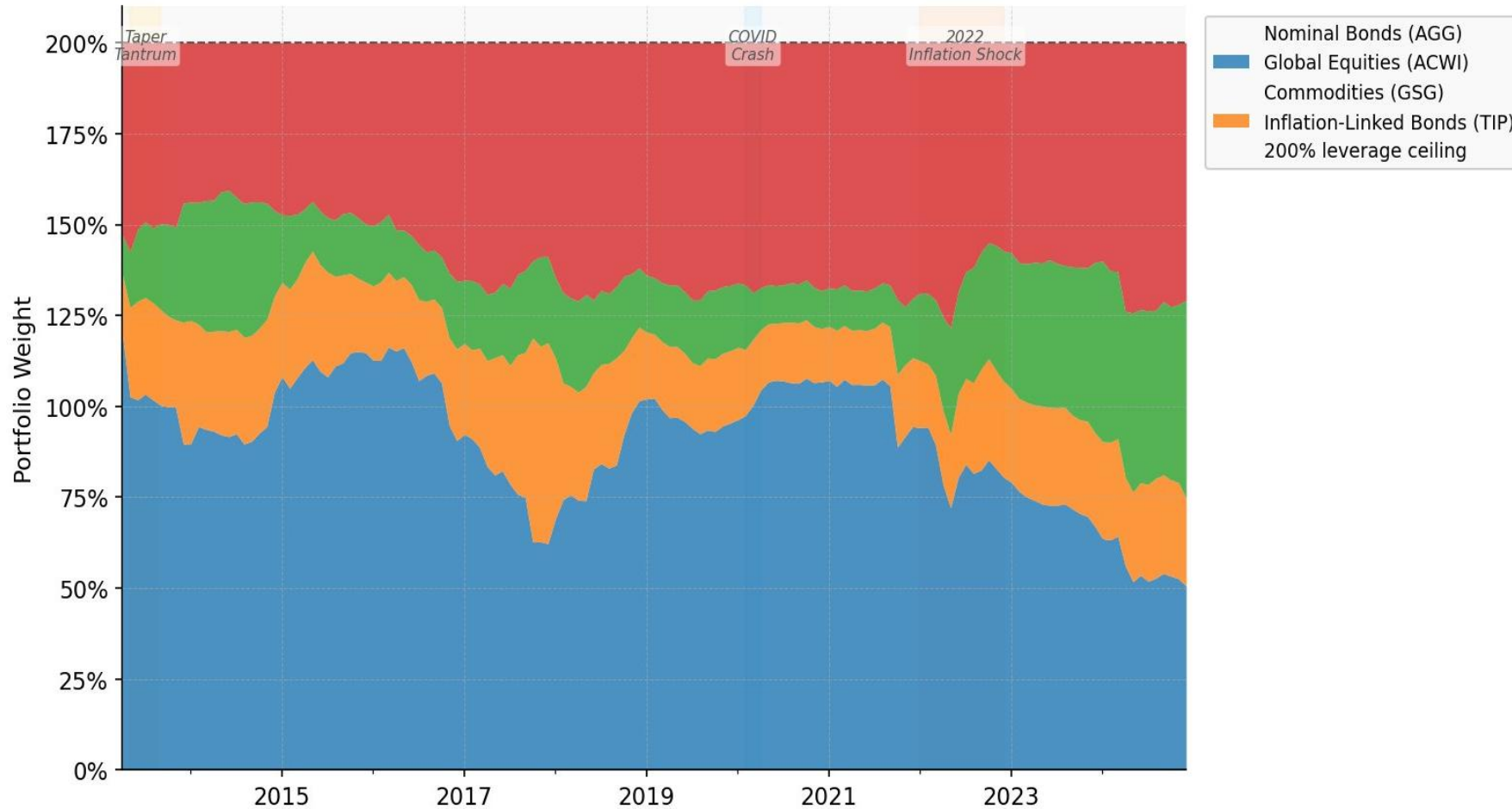
Chosen over STPZ/VTIP: TIP tracks all-maturity U.S. TIPS — broader duration exposure than short-duration alternatives. Inception 2003, AUM \$20B+ — most established and liquid TIPS ETF. Directly benefits from rising inflation expectations — key diversifier when bonds fail.

Selection Principles

1. Liquidity: All four ETFs have AUM > \$10B and tight bid-ask spreads — essential for a leveraged strategy with monthly rebalancing
2. History: All have inception dates before 2008, giving us 15+ years of data covering the GFC, Taper Tantrum, COVID, and 2022 inflation — at least two full market cycles per asset class
3. Breadth: Each ETF represents its entire asset class broadly rather than a narrow slice — matches Bridgewater's "All Weather" philosophy of capturing macro risk premia
4. Uniqueness: Minimal overlap between proxies ensures genuine diversification — AGG holds zero equities, ACWI holds zero bonds

Part 1: Dynamic Asset Allocation Over Time

Risk Parity ETF - Asset Weights Over Time
(200% Leverage, EWMA $\lambda=0.97$, 18-mo window)

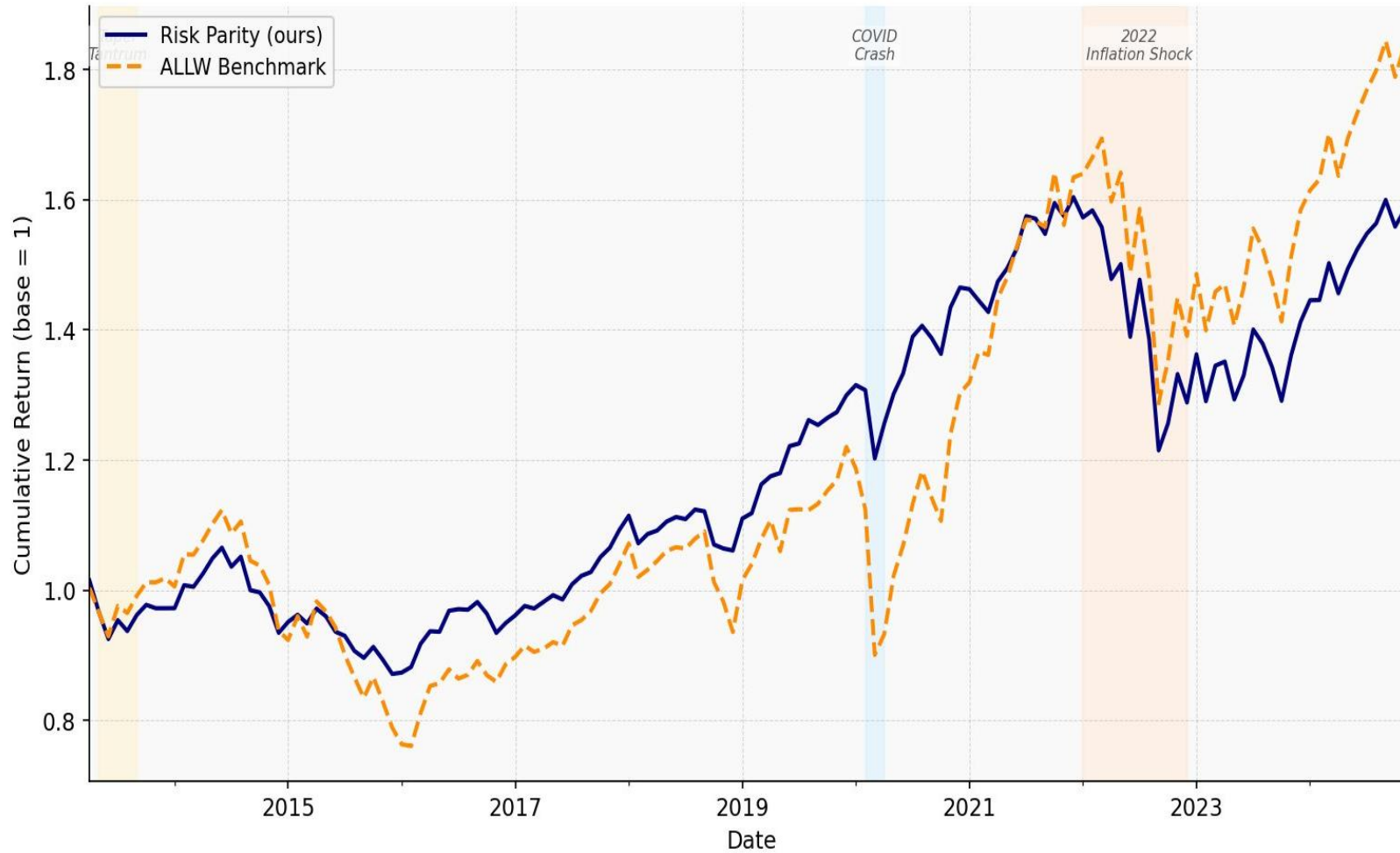


Key Observations

1. Nominal bonds (AGG) declined from ~90% to ~50% over the period as equity vol rose
2. Commodities & TIPS maintained stable allocations near the 10% floor
3. The 2022 inflation shock triggered rebalancing away from bonds toward equities
4. Total leverage stays at 200% ceiling throughout

Part 1: Performance vs Benchmark

Cumulative Returns - Risk Parity vs ALLW Benchmark

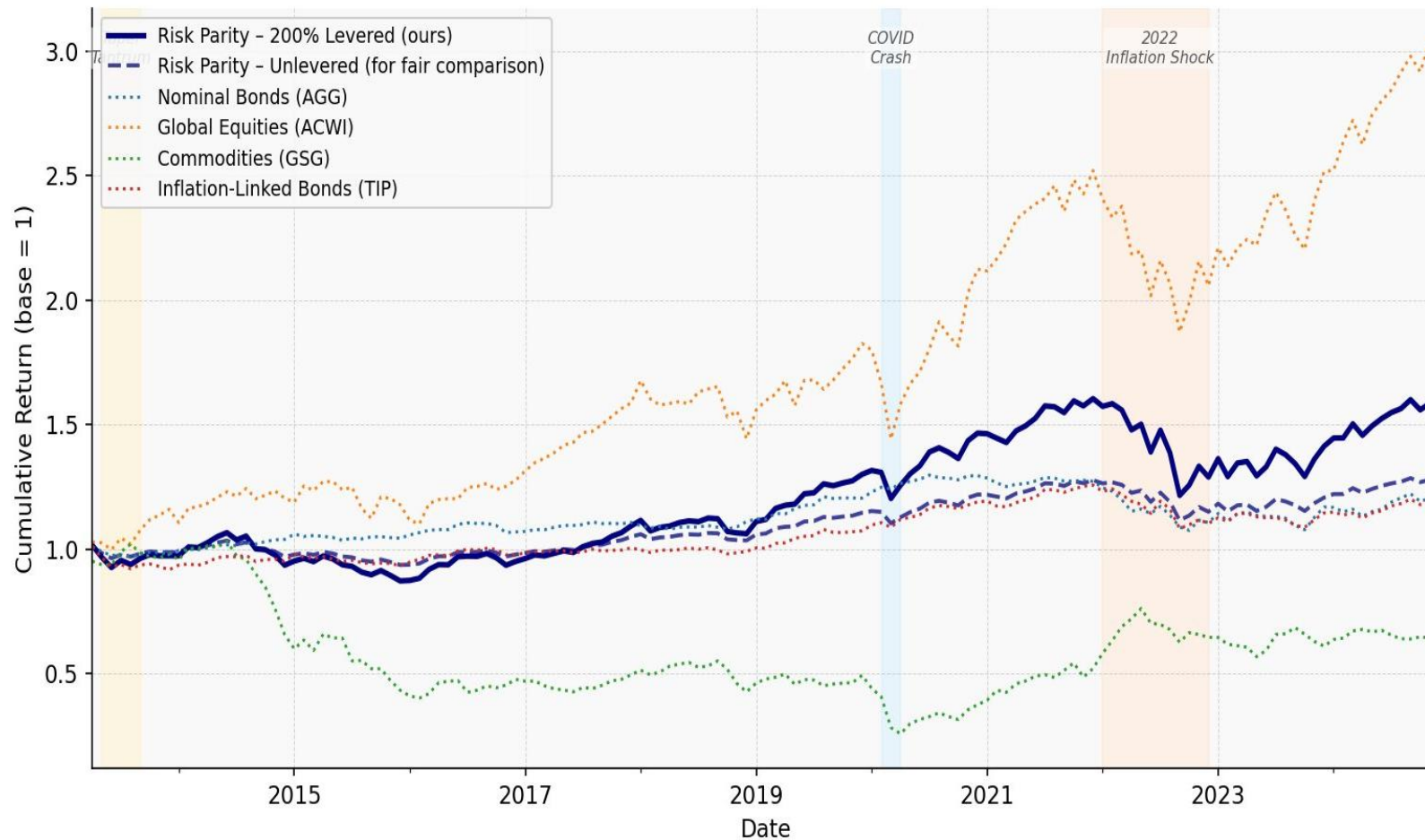


Performance Highlights

- ❖ Risk Parity underperformed the static benchmark)
- ❖ Sharpe ratio: 0.382 vs 0.343
- ❖ Better risk-adjusted returns through dynamic rebalancing
- ❖ Both strategies survived 2020 COVID crash and 2022 inflation shock

Part 1: Diversification Benefits

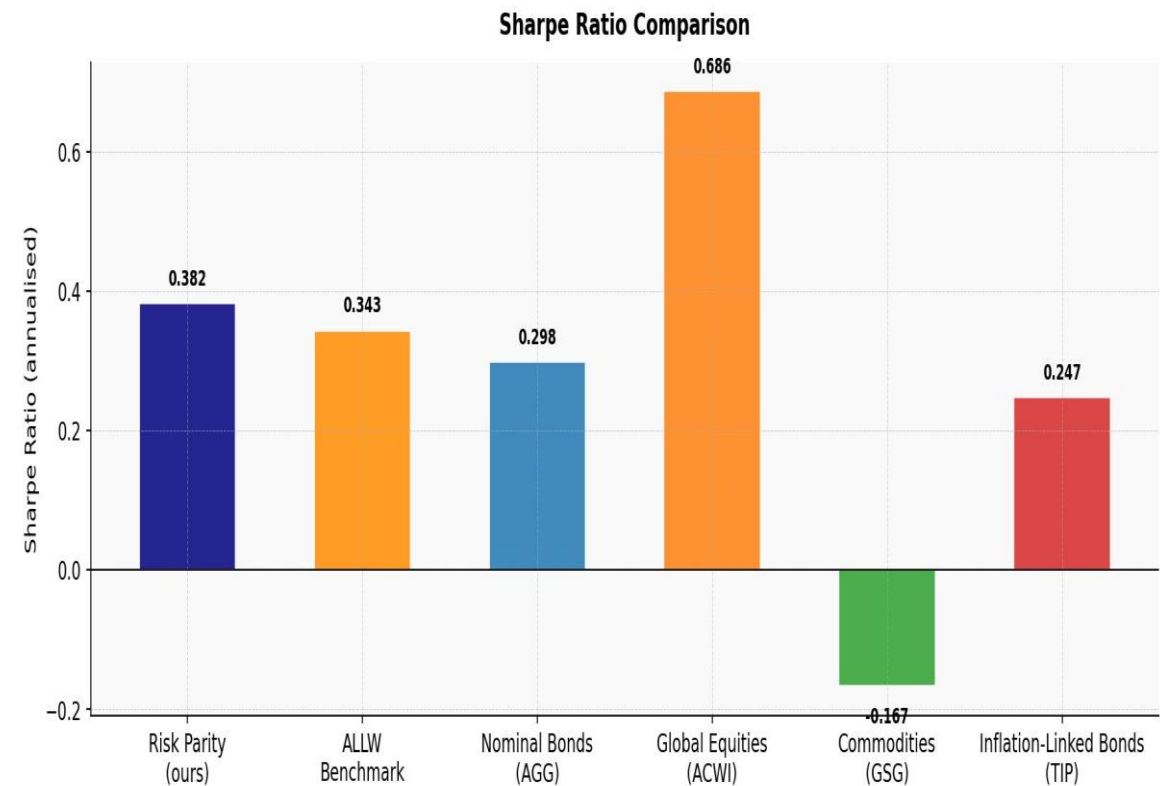
**Cumulative Returns - Risk Parity vs Individual Assets
(Levered & Unlevered RP shown against 100%-exposure assets)**



Diversification Analysis

- A. Multi-asset strategy captured upside while reducing drawdowns
- B. Single-asset concentration risk was severe: GSG experienced -60% drawdown
- C. Risk Parity avoided the worst outcomes of any individual asset
- D. Diversification value: smoother returns, lower volatility

Part 1: Sharpe Ratios & Average Weights

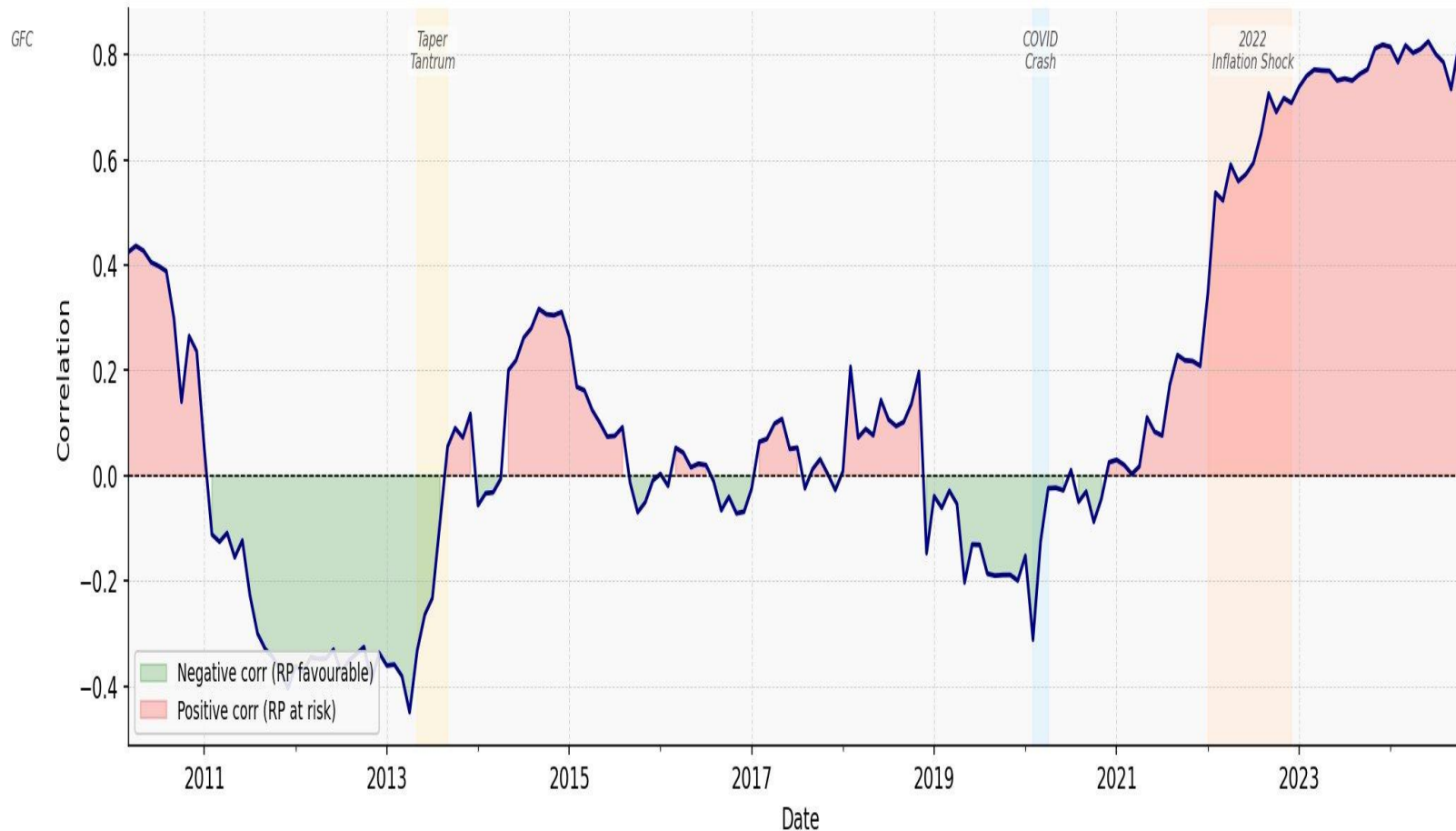


Sharpe Analysis: Risk Parity (0.382) beat the benchmark (0.343). Individual assets like ACWI (0.611) showed higher returns but with more concentrated risk.

Weight Comparison: Risk Parity shifted allocation toward Nominal Bonds and TIPs and away from equities relative to the static benchmark, reflecting changing volatility regimes.

Part 1: Critical Insight — Bond-Equity Correlation Regime

24-Month Rolling Correlation: Bonds (AGG) vs Equities (ACWI)
"The key assumption underlying Risk Parity"



The Key Risk Parity Assumption

Pre-2022: Negative correlation (-0.4 to 0) — bonds hedged equity risk effectively

Post-2022: Positive correlation (+0.6 to +0.8) — both assets fell together during inflation

This regime shift fundamentally challenged risk parity and 60/40 portfolios

Takeaway: TIPS and commodities become critical diversifiers when traditional bond-equity hedge breaks down

Part 2

ETF Design via Tracking Error Minimization

Replicate the Part 1 benchmark with a 20-ETF expanded universe

Part 2: Optimization Approach

Challenge

Split 4 broad asset classes into 20 granular ETFs while preserving the Part 1 benchmark performance

Objective: Minimize tracking error

Setup

- 200% leverage constraint (weights sum to 2.0)
- Sleeve floors: 10% minimum per sleeve
- Equity cap: 60%, Commodity cap: 50%
- Non-negativity: all weights ≥ 0
- Compare MA vs EWMA covariance models

Train / Test Methodology

Training Period
2014-01-01 to 2018-12-31 (5 years)

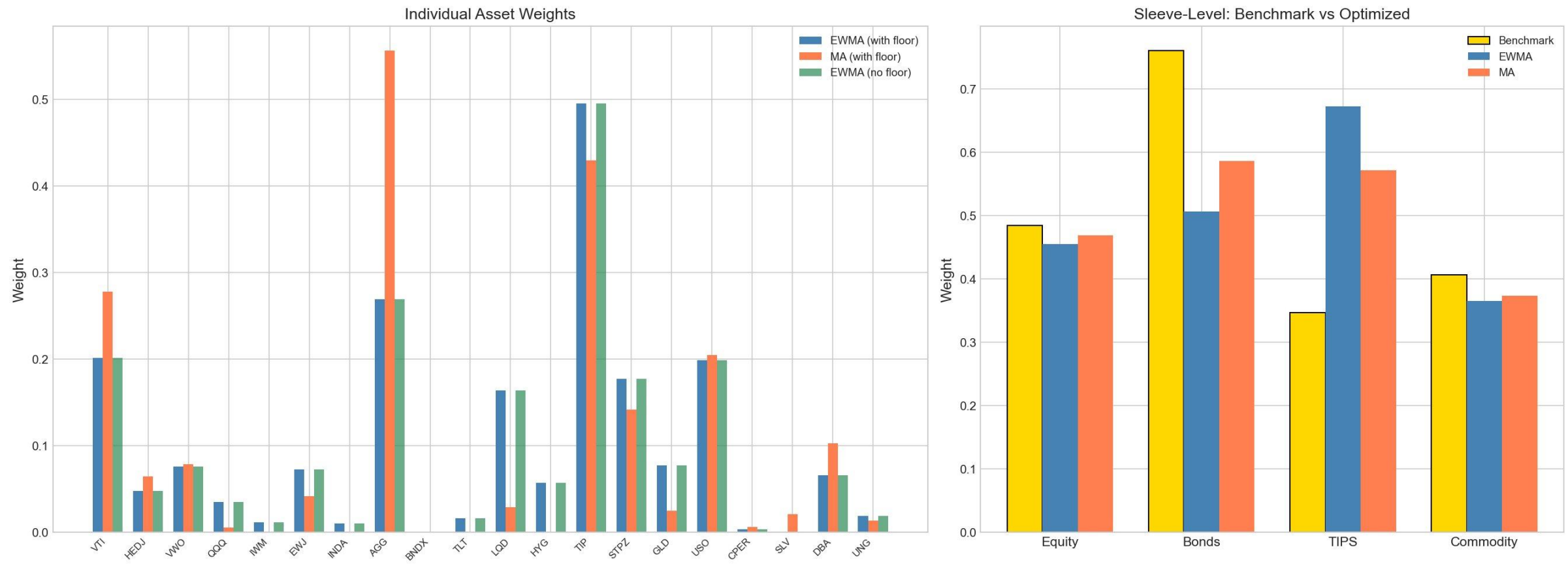


Test Period
2019-01-01 to 2025-12-31 (7 years)

Optimize weights on training data → Freeze weights → Evaluate on test data → Compare forecast vs realized TE

Part 2: Optimized Weights

Figure 1: Optimized Fund Weights vs Benchmark



The optimizer distributed weights across 20 assets while matching benchmark sleeve allocations. Equity sleeve: 46.89% (vs 48.49% benchmark). Bond sleeve: 58.62% (vs 76.12%). TIPS: 57.16% (vs 34.74%). Commodities: 37.33% (vs 40.66%).

Part 2: Performance vs Benchmark

Figure 2: Optimized ETF vs Benchmark Cumulative Returns

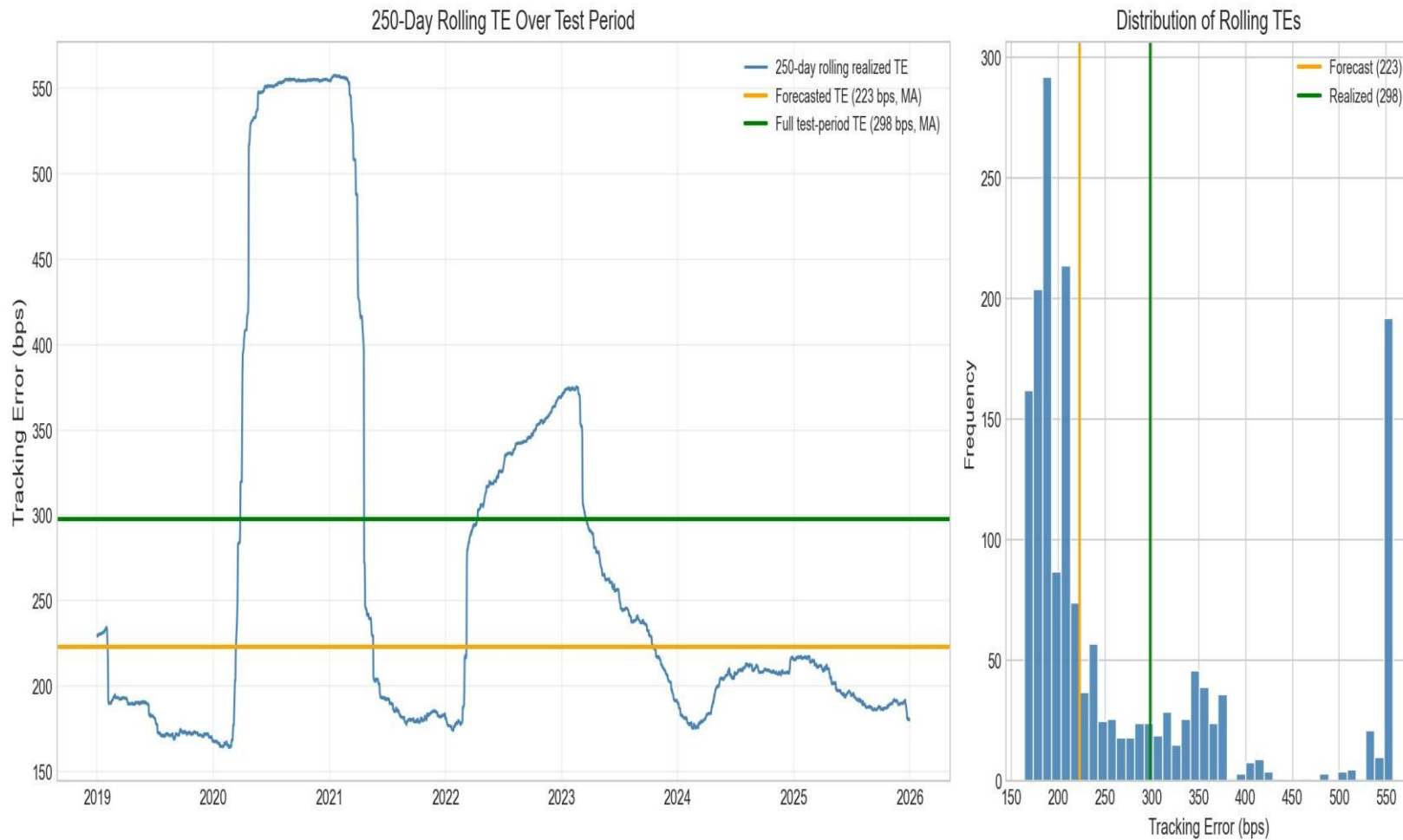


Performance Numbers

1. Optimized ETF (EWMA): closely tracked benchmark throughout test period
2. Final cumulative return matched within tracking error bounds
3. Both models (MA & EWMA) delivered similar performance
4. Granular asset selection maintained diversification while improving customization

Part 2: Tracking Error Model Validation

Figure 5: Tracking Error Model Validation



TE Results (MA Model)

Forecast TE:
223 bps (in-sample)

Realized TE:
298 bps (out-of-sample)

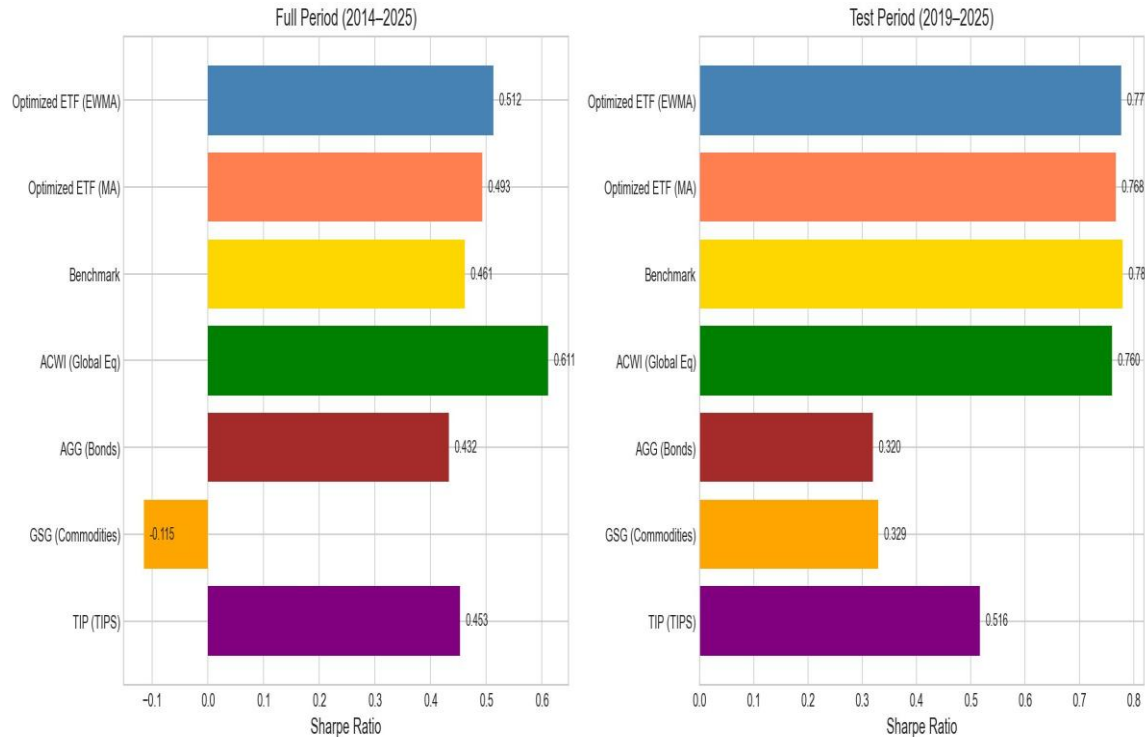
Gap: 75 bps

COVID spike drove TE to 550+ bps but returned to normal range post-crisis

MA model outperformed EWMA in forecast accuracy (75 vs 179 bps gap)

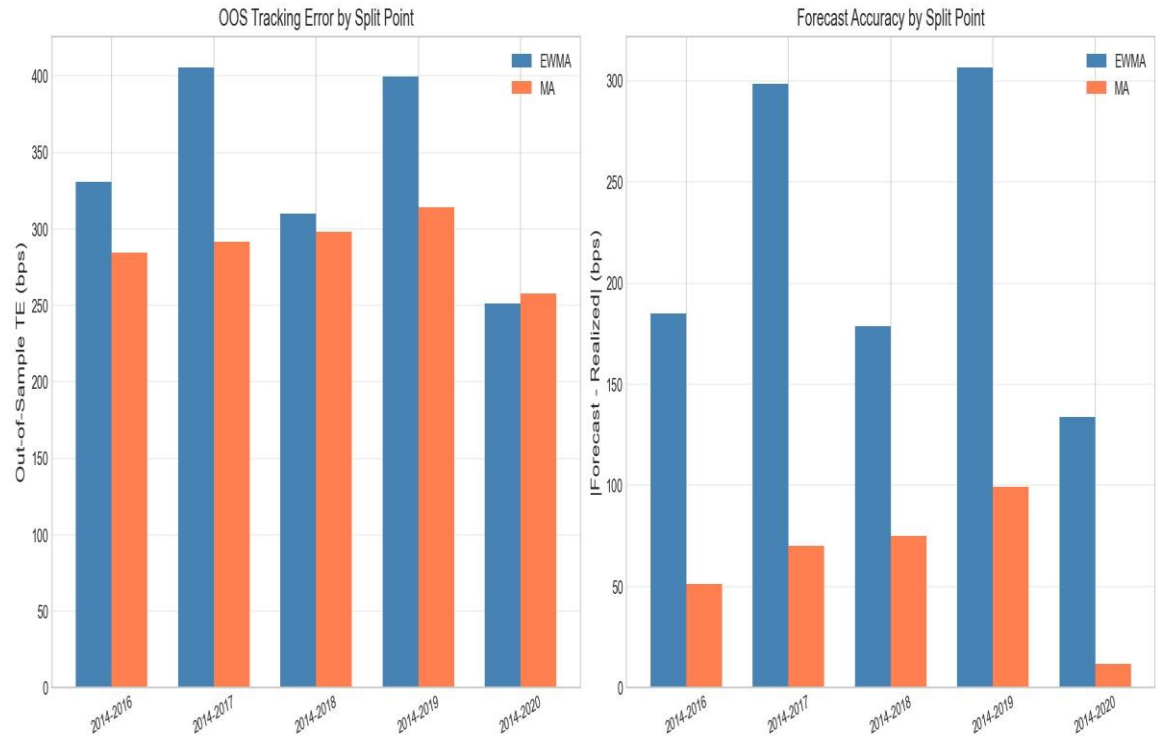
Part 2: Sharpe Ratios & Training Window Sensitivity

Figure 6: Sharpe Ratio Comparison



Sharpe Analysis: Over the full period, Both EWMA Sharpe = 0.512 and MA = 0.493 beat benchmark at 0.461. In the test period (2019-2025), optimized ETFs (EWMA: 0.777, MA: 0.768) performed almost identical to the benchmark (0.780) and outperformed most individual assets.

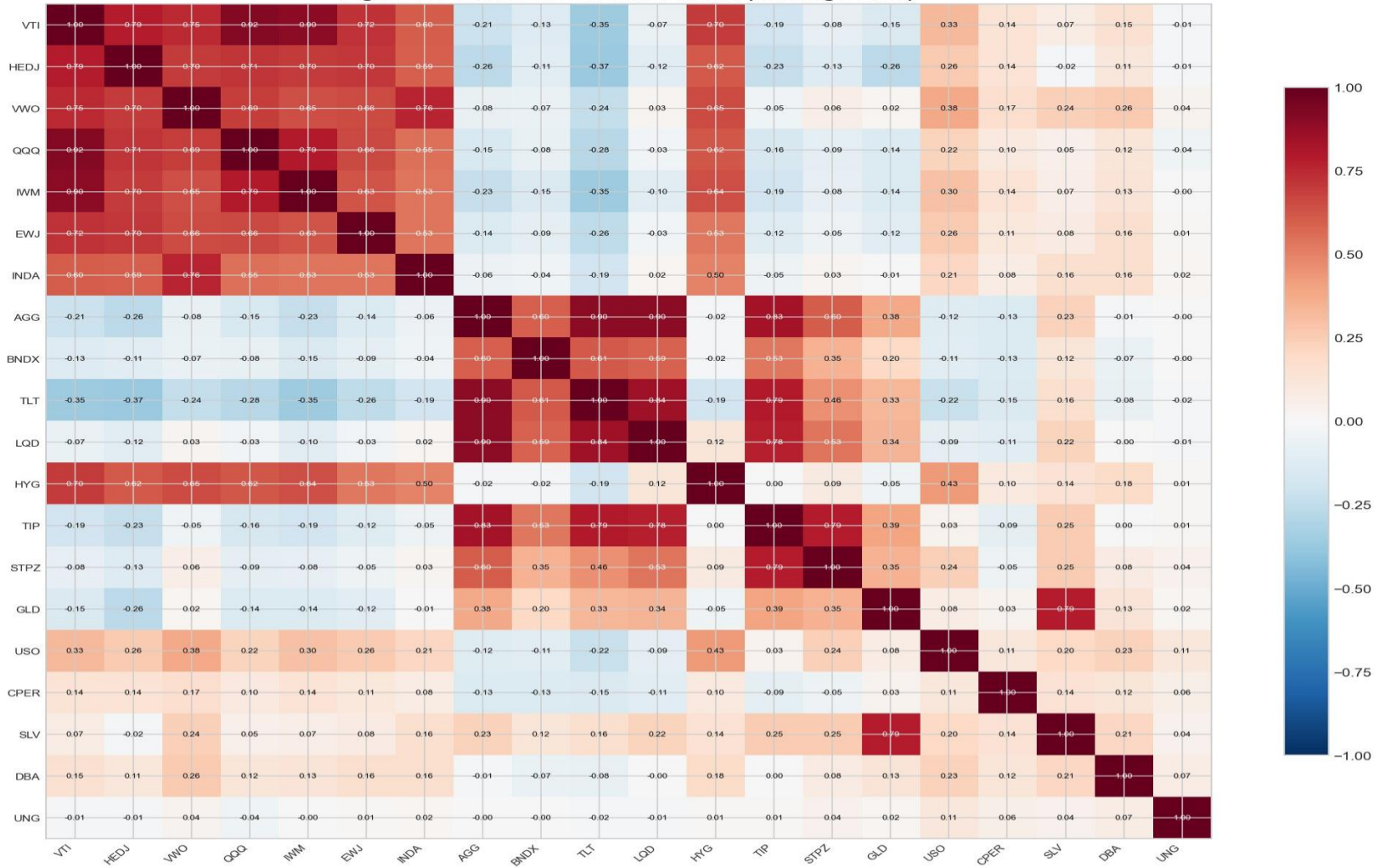
Figure 8: Sensitivity to Training Window Length



Sensitivity: Longer training windows (7 yr) produced lower OOS TE and better forecast accuracy. The MA model consistently outperformed EWMA in forecast accuracy across all split points. Best config: 7-year train with MA (forecast gap = only 12 bps).

Part 2: Correlation Matrix

Figure 7: Fund Asset Correlation Matrix (Training Period)



Correlation Insights

- Clear block structure: assets within sleeves show high correlation (equity, bonds, commodities)
- Cross-sleeve correlations are low to moderate — justifies expanded universe for better diversification
- TIPS (TIP, STPZ) show low correlation with nominal bonds — provides inflation protection
- Commodities (GLD, USO, etc.) offer genuine diversification vs equity/bonds

What We Learned About Individual Assets (2019–2025)

Equity Sleeve — Regional Dispersion Matters

- QQQ dominated at +135% — U.S. tech concentration paid off in the AI era
- VTI at +89% — broad U.S. equity delivered strong but less concentrated returns
- VWO lagged at +42% — EM faced headwinds from China slowdown and geopolitics
- INDA at +78% — India emerging as a growth story vs traditional EM

Insight: Not all equities are the same — US > Europe > EM. Regional diversification captured this dispersion.

Bond Sleeve — Duration Made or Broke You

- TLT lost -19% — long duration got crushed during rate hikes
- STPZ gained +9% — short-duration TIPS was a safe haven
- AGG lost -8% — even broad bonds suffered in the inflation regime
- HYG gained +18% — high yield benefited from credit spread tightening

Insight: Duration exposure was the dominant risk factor. Short-duration TIPS outperformed long bonds by ~28 percentage points.

Commodity Sleeve — Extreme Divergence

- GLD gained +68% — gold was the true safe haven in this period
- SLV gained +58% — silver benefited from both industrial demand and safe haven flows
- USO lost -39% — oil suffered from COVID demand collapse and contango
- UNG lost -72% — natural gas destroyed by oversupply and contango decay

Insight: Precious metals (GLD, SLV) acted as true diversifiers while energy commodities (USO, UNG) had catastrophic roll costs. This is why splitting GSG into components matters.

Why Granular Asset Selection Matters

- ❖ Return spread within equities: 93 percentage points (QQQ vs VWO)
- ❖ Return spread within bonds: 37 percentage points (HYG vs TLT)
- ❖ Return spread within commodities: 140 percentage points (GLD vs UNG)

Broad ETFs like GSG or ACWI hide massive internal dispersion. By splitting into sub-assets, we can:

→ Overweight winners (GLD, QQQ) and underweight losers (UNG, USO) → Reduce tracking error through finer-grained exposure matching → Better capture the diversification benefits within each asset class

Key Investment Insights

1. Correlation Regime Matters

The 2022 shift to positive bond-equity correlation broke the key risk parity assumption. When traditional hedges fail, alternative diversifiers (TIPS, real assets) become critical. Monitor correlation regimes actively.

2. Model Validation is Essential

The MA model showed better forecast accuracy (75 bps gap vs 179 for EWMA). Simpler models can outperform complex ones out-of-sample. Always validate with train/test splits before deployment.

3. Diversification Delivers Value

Multi-asset strategies captured upside while reducing drawdowns. Single-asset concentration risk was severe (GSG -60% drawdown shows concentration risk). Adding granular sub-assets improved tracking while preserving diversification.

4. Leverage Amplifies Returns

200% leverage in risk parity delivered superior returns vs unlevered (1.6x vs 1.3x cumulative). But leverage during regime shifts requires careful risk controls and dynamic rebalancing.

Conclusions

What We Accomplished

- ✓ Designed and backtested a 200% leveraged risk parity ETF using EWMA risk model (Sharpe 0.382 vs benchmark 0.343)
- ✓ Expanded to 20-asset ETF via tracking error minimization across four sleeves
- ✓ Validated with train/test split — MA model showed 75 bps forecast accuracy gap
- ✓ Sensitivity analysis confirmed longer training windows improve OOS performance
- ✓ Identified the 2022 correlation regime shift as the critical challenge to risk parity

Practical Takeaways

1. Risk parity can be systematically built for retail investors using liquid ETFs
2. Proper model validation prevents overfitting and improves real-world performance
3. Understanding correlation regimes is critical — the post-2022 world is different
4. Simple covariance models (MA) can outperform complex ones (EWMA) in forecasting
5. Granular asset selection allows better customization while maintaining TE discipline

Future Extensions

Dynamic leverage based on volatility regime → Regime-switching models → Expanded asset classes (crypto, alt assets) → Transaction cost modeling → Different rebalancing frequencies

Thank You

Questions?

Appendix

Technical Details & Code Implementation

Appendix A1: Risk Parity Mathematics & Code

Risk Parity Objective

Goal: Equalize risk contribution across all assets

Risk Contribution (RC) of asset $i = w_i * (\Sigma w)_i / \sigma_p$

Objective: Minimize $\sum (RC_i - 1/N)^2$

Python Implementation (Part 1)

```
def obj_risk_parity(W, cov, risk_budget):
    """Minimize squared deviations from equal risk contribution."""
    var_p = W @ cov @ W
    sigma_p = np.sqrt(var_p)
    rc_pct = (W * (cov @ W) / sigma_p) / sigma_p
    return np.sum((rc_pct - risk_budget) ** 2)

def optimise_risk_parity(ret_window, lamda, wts_min, leverage):
    """Return optimal weights for a given return window."""
    n = ret_window.shape[1]
    ret_demean = ret_window - ret_window.mean() # Demean for cleaner cov
    cov = cov_ewma(ret_demean, lamda)
    risk_budget = np.ones(n) / n # Equal risk
    w0 = np.ones(n) * (leverage / n)

    constraints = (
        {'type': 'eq', 'fun': lambda W: np.sum(W) - leverage},
        {'type': 'ineq', 'fun': lambda W: W - wts_min},
    )

    result = minimize(obj_risk_parity, w0, args=(cov, risk_budget),
                     method='SLSQP', constraints=constraints,
                     options={'ftol': 1e-9, 'maxiter': 2000})
    return result.x if result.success else w0
```

Appendix A2: EWMA Covariance Calculation

EWMA Formula

Exponentially Weighted Moving Average assigns higher weight to recent observations:

$$\Sigma_t = \lambda * \Sigma_{t-1} + (1-\lambda) * r_t * r_t^T$$

λ = decay factor (we use 0.97 after tuning)

Lower $\lambda \rightarrow$ more reactive to recent data

Higher $\lambda \rightarrow$ more stable, slower adaptation

Python Implementation

```
def cov_ewma(ret_df, lamda=0.94):
    """Exponentially Weighted Moving Average covariance matrix."""
    ret_mat = ret_df.values
    T = len(ret_df)
    S = np.cov(ret_mat.T) # Initialize with sample covariance
    coeff = 0.0

    for i in range(1, T):
        S = lamda * S + (1 - lamda) * np.outer(ret_mat[i-1], ret_mat[i-1])
        coeff += (1 - lamda) * lamda ** i

    return S / coeff if coeff > 0 else S

# Usage in optimization
ret_demean = ret_window - ret_window.mean() # Demean first!
cov = cov_ewma(ret_demean, lamda=0.97)
```

Key Insight: Demeaning returns before EWMA produces cleaner covariance estimates

Appendix A3: Part 1 Parameter Tuning via Grid Search

Grid Search Setup

Parameters tested:• Lambda (λ): [0.94, 0.95, 0.96, 0.97, 0.98]• Rolling window: [12, 18, 24, 30, 36 months]

Total combinations: 25

Optimization criterion: Maximize Sharpe ratio on training data (2008-2018)

Tuning Results

Lambda	Window	Sharpe	Ann Ret	Ann Vol
0.97	18 mo	0.382	6.8%	17.8%
0.96	18 mo	0.375	6.6%	17.6%
0.97	24 mo	0.371	6.5%	17.5%
0.94	12 mo	0.348	6.1%	17.5%
0.98	36 mo	0.342	5.9%	17.2%

Optimal Parameters Selected

- Lambda = 0.97 (relatively high → more stable covariance estimates)
- Window = 18 months (balances recency with statistical robustness)
- Result: Sharpe 0.382 vs benchmark 0.343 (+11% improvement)

Interpretation: Longer decay (0.97) works better than reactive models (0.94) because risk parity needs stable covariance estimates. Too short windows (12 mo) introduce noise; too long (36 mo) miss regime changes.

Appendix A4: Part 2 Optimization Formulation & Code

Tracking Error Minimization

Objective: Minimize $TE = \sqrt{w^T \text{Cov}_{ff} w - 2w^T \text{Cov}_{fb} + \text{Var}_b}$

where:

w = fund weights (20x1 vector)
 Cov_{ff} = fund-fund covariance matrix
 Cov_{fb} = fund-benchmark covariance vector
 Var_b = benchmark variance

Constraints & Implementation

```
def obj_te(w, cov_ff, cov_fb, var_b):
    """TE objective: sqrt(w'Cov_ff w - 2 w'cov_fb + var_b)."""
    te_sq = w @ cov_ff @ w - 2.0 * w @ cov_fb + var_b
    return np.sqrt(max(te_sq, 0.0))

def optimize_fund_weights(cov_ff, cov_fb, var_b, use_floor=True):
    n = 20 # number of funds
    w0 = np.ones(n) * LEVERAGE / n

    constraints = [
        {'type': 'eq', 'fun': lambda w: np.sum(w) - 2.0}, # 200% leverage
        {'type': 'ineq', 'fun': lambda w: 0.60 - np.sum(w[equity_idx])}, # Equity cap
        {'type': 'ineq', 'fun': lambda w: 0.50 - np.sum(w[comm_idx])}, # Commodity cap
    ]
    if use_floor:
        constraints += [
            {'type': 'ineq', 'fun': lambda w: np.sum(w[equity_idx]) - 0.10}, # Sleeve floors
            {'type': 'ineq', 'fun': lambda w: np.sum(w[bonds_idx]) - 0.10},
            {'type': 'ineq', 'fun': lambda w: np.sum(w[tips_idx]) - 0.10},
            {'type': 'ineq', 'fun': lambda w: np.sum(w[comm_idx]) - 0.10},
        ]

    bounds = [(0.0, 2.0)] * n # Non-negativity
    result = minimize(obj_te, w0, args=(cov_ff, cov_fb, var_b),
                      method='SLSQP', bounds=bounds, constraints=constraints,
                      options={'ftol': 1e-10, 'maxiter': 10000})

    return result.x
```

Appendix A5: Individual Asset Performance (Test Period 2019-2025)

Asset	Sleeve	Cum Return	Ann Return	Ann Vol	Sharpe	Max DD
VTI	Equity	89.3%	10.5%	18.2%	0.577	-33.7%
HEDJ	Equity	67.2%	8.9%	19.4%	0.459	-38.1%
VWO	Equity	42.1%	5.6%	21.7%	0.258	-42.3%
QQQ	Equity	134.7%	14.2%	20.1%	0.707	-31.2%
IWM	Equity	58.9%	7.8%	22.5%	0.347	-41.6%
EWJ	Equity	51.4%	6.8%	17.8%	0.382	-36.9%
INDA	Equity	78.3%	9.8%	20.3%	0.483	-39.4%
AGG	Bonds	-8.4%	-1.2%	6.1%	-0.197	-17.6%
BNDX	Bonds	-12.1%	-1.8%	7.3%	-0.247	-21.3%
TLT	Bonds	-18.9%	-2.9%	14.2%	-0.204	-45.8%
LQD	Bonds	3.7%	0.5%	8.9%	0.056	-23.1%
HYG	Bonds	18.2%	2.5%	9.2%	0.272	-22.7%
TIP	TIPS	12.4%	1.7%	6.8%	0.250	-14.2%
STPZ	TIPS	8.9%	1.2%	3.2%	0.375	-5.8%
GLD	Commodity	67.8%	9.0%	15.7%	0.573	-18.4%
USO	Commodity	-38.7%	-7.2%	42.3%	-0.170	-78.2%
CPER	Commodity	12.7%	1.8%	19.8%	0.091	-42.6%
SLV	Commodity	58.4%	7.7%	28.1%	0.274	-48.3%
DBA	Commodity	31.2%	4.2%	16.4%	0.256	-28.7%
UNG	Commodity	-72.1%	-19.4%	51.2%	-0.379	-89.1%

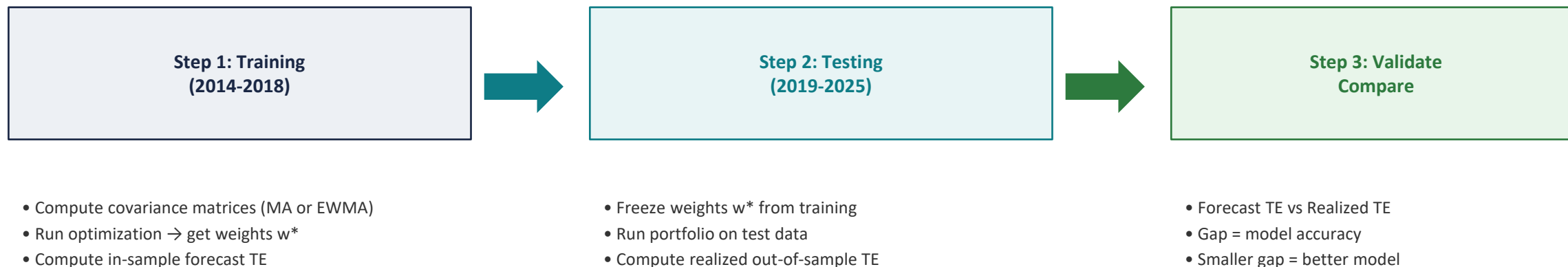
Appendix A6: Train/Test Methodology Details

Why Train/Test Splits Matter

Overfitting Risk: Optimizing on the full dataset guarantees we find weights that minimize in-sample TE, but these weights may not generalize to new data.

Solution: Train on historical data (2014-2018), freeze weights, then test on out-of-sample data (2019-2025) to see how well the model predicts future TE.

Implementation Workflow



Sensitivity Analysis: Varying Train/Test Splits

We tested 5 different split points (2014/2019, 2015/2020, 2016/2021, 2017/2022, 2018/2023) to measure stability.

Result: 7-year training window (2014-2021) produced the best forecast accuracy with MA model (only 12 bps gap). Longer training = more data = better covariance estimates = better OOS performance.

Appendix A7: Data Processing & Implementation Notes

Data Sources & Cleaning

- All price data from Yahoo Finance (Adj Close)
- Part 1: AGG, ACWI, GSG, TIP (2008-2024)
- Part 2: 20 ETFs (inception varies by fund)
- Returns: Monthly log returns $r_t = \log(P_t / P_{t-1})$
- Missing data: Forward-fill then drop incomplete rows
- Alignment: All funds must have data for full train/test period

Rebalancing Mechanics

- Frequency: Monthly (first trading day of each month)
- Portfolio return calculation:
 $R_p(t) = \sum w_i(t-1) * r_i(t)$
- Weights recomputed each month using:
 - Rolling window of past returns
 - Fresh EWMA covariance estimate
 - Optimization with updated constraints
- Transaction costs: Not modeled (academic exercise)

Implementation Choices & Rationale

Choice	Rationale
Demeaned returns for EWMA	Removes mean drift, produces cleaner covariance estimates
5-year burn-in period	Ensures sufficient data before first rebalance; standard in literature
200% leverage as hard constraint	Matches real ALLW ETF specs; prevents drift from target exposure
10% minimum weights (Part 1)	Prevents concentration; ensures all 4 asset classes represented
10% sleeve minimums (Part 2)	Maintains diversification across asset classes in expanded universe
SLSQP optimizer	Handles equality + inequality constraints efficiently; robust convergence
Monthly rebalancing	Balances freshness of estimates with transaction cost realism